
Advertising Revenue & Sales Performance

Revenue is growing 9.9% — so why is that a problem?

Yizhou Liu · Data Analyst · Sydney, Australia

SQL (DuckDB — CTEs, window functions, cohort analysis) · Python (pandas, matplotlib) · Tableau

github.com/liuyizhou0402/data-analysis / [ads-revenue-analytics](#)

\$566.3M

revenue analysed

340

advertisers

195,488

campaign-days

12

SQL business queries

An end-to-end business analysis of a digital advertising sales organisation over 18 months — 340 advertisers, 42 sales reps, 3,143 campaigns. Twelve SQL queries run in the order a monthly business review runs: *what happened* → *why* → *who* → *what next*. The project demonstrates one analytical habit: the headline number said the business was healthy, and the job was to check whether that was true.

Four headline findings

- **Growth is masking contraction.** Revenue rose 9.9% half-over-half while active advertisers fell **31%** and revenue per advertiser rose **93%** — expansion on a shrinking base.
- **One segment is in structural decline, invisible in the trend.** SEA Gaming contracted **33.7%**, and an independent churn query returns 5 SEA Gaming accounts in its top 10 at-risk list.
- **Book size measurably damages retention.** Reps with 17+ accounts lose **71%** of their book versus 43% for reps with 1–8 — and the effect survives controlling for tier mix.
- **The highest-return vertical is under-invested.** Beauty & Personal Care returns **4.75x ROAS** on 10.7% of spend, against a 2.95x book average.

On the data

The dataset is **synthetic and seeded** (SEED = 20260218) — advertiser-level revenue is commercially confidential and never public. The generator builds in six business patterns for the analysis to find rather than producing noise. Getting those patterns to survive into the data took more work than generating it did; see the appendix.

The headline is hiding two problems

Revenue grew **9.9%** half-over-half — \$184.0M in the first six months to \$202.2M in the last six. On that number alone the business looks healthy. It is not the whole picture.

	First month	Last month	Change
Active advertisers	199	138	−31%
Revenue per advertiser	\$144,681	\$278,565	+93%

(Q1 — executive summary)

All of the growth is coming from existing accounts spending more, on a base that is contracting by roughly a third. ARPA nearly doubling is not a success story on its own — it is what revenue concentration looks like while it is happening. The top revenue decile (31 advertisers) now carries **66.8%** of revenue, and the top two deciles carry **83.2%** (Q6).

Why it matters

That is the single most important thing in this report, and no single-number dashboard would have surfaced it. Blended growth averaged a rising E-commerce book against a falling Gaming one and reported the difference as health. Two specific problems sit underneath it — a declining segment (\$1) and broken retention in the tiers that feed the base (\$2).

Summary dashboard

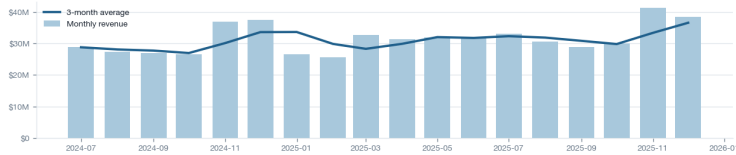
Rendered from the query extracts by `analysis/build_dashboard.py`. Twelve Tableau-ready CSVs, a step-by-step build guide and every calculated field (with its reasoning) ship in `dashboard/` so the interactive workbook reproduces from the same source.

Advertising Revenue & Sales Performance

Jul 2024 – Dec 2025 · 18 months · \$566M revenue · 3.30x blended ROAS · synthetic data

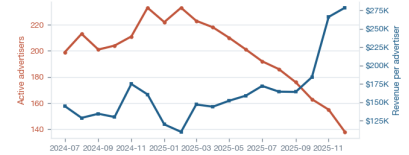
Revenue is growing — but the growth is not evenly distributed

Q4 years in both years, the last 3-month average through mid-2025 is the real trend



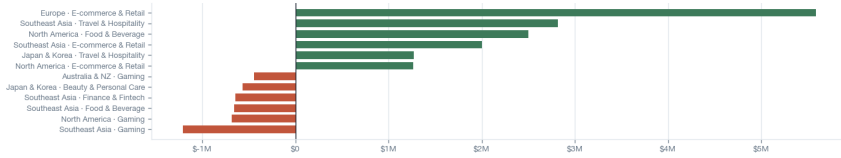
The account base is shrinking

Growth is coming from lower, larger accounts



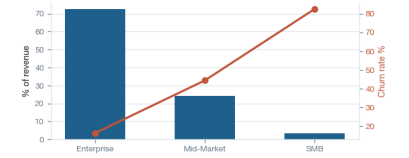
Revenue diagnosis: last 3 months vs prior 3 months

Southeast Asia Gaming is the single largest drag — invisible in the headline trend



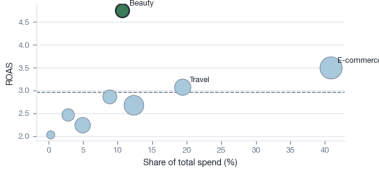
Churn is concentrated where revenue isn't

SMB: 83% churn, 3% of revenue



Where return is high but investment isn't

Beauty: best ROAS, only 11% of spend



New reps take ~5 months to ramp

Reaching 91% of a tenured rep's monthly revenue by month 4



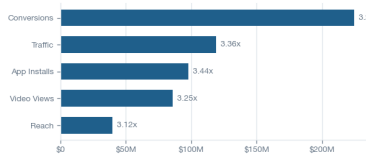
Bigger books, worse retention

ROAS after controlling for account size mix



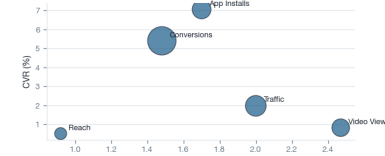
Spend by campaign objective

Labels show ROAS



Click rate vs conversion rate

Bubble size = share of spend, the two trade off



Returns are stable while volume moves

ROAS held 3.17x-3.37x across the period



1. Gaming is in structural decline, and it is being masked

Comparing the most recent three months against the prior three, total revenue rose \$17.2M. Within that, declining segments removed \$4.9M — and the largest single drag is Southeast Asia Gaming: **−\$1.21M, −33.7%**, taking **7.0 percentage points** off the total change on its own (Q3). Gaming declines are not confined to one market:

Segment	Change	% change
Southeast Asia · Gaming	−\$1,205,536	−33.7%
North America · Gaming	−\$685,522	−31.7%
Australia & NZ · Gaming	−\$444,877	−65.4%
Japan & Korea · Gaming	−\$180,646	−14.5%
Europe · Gaming	growth	+47.1%

The check that makes it credible

Corroborated by a query that did not know about it. The churn watchlist (Q12) was built independently of the segment analysis and returns 10 at-risk accounts carrying \$2.84M — **6 of the 10 are Gaming advertisers, 5 of them in Southeast Asia**. Two unrelated cuts of the data landing on the same root cause is a far stronger signal than either alone, and it is the reason this finding is reported as a conclusion rather than a hypothesis.

Recommendation

- Treat the Gaming book as a **retention** problem, not a performance one — Gaming ROAS is 2.68x against a 2.95x vertical average, so advertisers are getting weaker returns and responding rationally.
- Commission a competitive review specifically for SEA Gaming **before** next quarter's targets are set. A –34% segment trend that continues removes roughly **\$2.4M annualised**.
- Do not set regional targets from the blended growth rate — it will over-target exactly the markets where Gaming concentrates.

2. The account base is shrinking because SMB retention is broken

Churn is severe and heavily tiered (Q5):

Tier	Advertisers	% of revenue	Churn rate	Revenue / advertiser
Enterprise	42	72.5%	16.3%	\$9,770,015
Mid-Market	103	24.2%	44.4%	\$1,329,444
SMB	162	3.4%	82.5%	\$117,601

SMB is 48% of the account base and 3.4% of revenue, churning at 82.5%.

Recommendation — and a caution

The obvious read is wrong. "Fix SMB retention" is the natural response, and the number that kills it is the third column: SMB is **3.4% of revenue**. A programme that halves SMB churn is worth about \$1.7M annually in the best case and would consume disproportionate service capacity. The better use of the same finding is **coverage design** — move SMB to pooled or self-serve, freeing named reps for Mid-Market, where churn is still 44.4% but each account is worth **11x more**. Mid-Market is where retention spend earns its return.

3. Rep book sizes are set too high, and it is costing retention

Churn on a rep's book rises with the size of that book (Q9):

Book size	Reps	Avg book	Book churn	SMB-only churn
1–8 accounts	14	5.1	42.8%	65.0%
9–16 accounts	8	11.5	56.1%	75.0%
17+ accounts	5	30.6	71.3%	93.8%

The relationship survives the obvious objection. Reps carrying more accounts might simply hold more SMB accounts, which churn more for unrelated reasons — a confound that would make the finding meaningless. The SMB-only column controls for exactly that, and the effect is *stronger* within SMB alone (65.0% → 93.8%) than across all accounts. Book size is doing the work, not tier mix. Reps above the line hold **38% of total revenue**, so this is not a marginal group.

Recommendation

Cap named-coverage books at roughly **16 accounts**. The five reps above that line average 30.6 accounts and lose 71% of their book. Rebalancing requires either redistributing accounts to reps below the line or moving their SMB tail to pooled coverage — which is the same action recommended in §2. Doing both together is what makes either affordable.

4. Beauty & Personal Care is under-invested

Beauty & Personal Care returns **4.75x ROAS** — the highest of any vertical, against a 2.95x average — on only **10.7% of spend**, across 29 advertisers generating \$60.4M (Q11). For comparison, E-commerce & Retail takes 40.8% of spend at 3.50x.

- Set an upsell target on the existing 29 Beauty accounts **before** pursuing new logos — expanding a book that already returns 4.75x is cheaper than acquiring into it.
- Prioritise Beauty in acquisition for the next two quarters, particularly in Europe, where the vertical grew 21.1% and is not yet saturated.

Caveat

ROAS here is attributed, not incremental. High attributed ROAS can partly reflect purchases that would have happened anyway, and beauty is a category where that risk is real. The upsell recommendation is safe either way; a large acquisition bet should be validated with an incrementality test first. Stating this is not hedging — it is the difference between a recommendation a business can act on and one that quietly transfers risk to whoever acts on it.

5. New reps take five months to ramp — plan hiring accordingly

Reps hired inside the observation window reach (Q8):

Months since hire	0	1	2	3	4	5
% of tenured productivity	25%	49%	54%	76%	91%	93%

15 of 38 reps were hired during the window, so this is measured on a substantial group rather than a handful. Hiring decisions need a five-month lead time against revenue targets: headcount to support a Q4 peak must be hired by **Q2, not Q3**. And given §3, the accounts freed by capping book sizes should go to *tenured* reps, not to new hires.

Summary of recommended actions

#	Action	Evidence	Expected impact
1	Competitive review of SEA Gaming before next targets are set	Q3, Q12	Protects ~\$2.4M annualised
2	Move SMB to pooled / self-serve coverage	Q5, Q9	Frees capacity; SMB is 3.4% of revenue
3	Cap named books at ~16 accounts	Q9	Addresses 38% of revenue held by over-loaded reps
4	Upsell the 29 existing Beauty accounts	Q11	Highest-ROAS vertical, on 10.7% of spend
5	Hire five months ahead of target periods	Q8	Removes the ramp gap from capacity planning
6	Report growth by segment, not blended	Q1, Q3	Blended growth is currently masking a –34% segment

Appendix — how it was built

The twelve queries

Ordered the way a monthly business review runs — *what happened* → *why* → *who* → *what next*.

	Question	Techniques
Q1	Executive KPI summary with MoM movement	CTEs, LAG, guarded rate maths
Q2	Revenue trend, moving average, running total	AVG OVER with explicit frame, SUM OVER
Q3	Revenue diagnosis — which segments moved	Conditional aggregation, contribution-to-change
Q4	Advertiser cohort retention	Cohort self-join, DATE_DIFF, pivot via CASE
Q5	Retention and revenue by account tier	Multi-CTE join, share-of-total window
Q6	Advertiser value deciles and Pareto curve	NTILE, running total window
Q7	Sales rep scorecard	RANK, partitioned rank, MEDIAN as window
Q8	New-hire ramp curve	Date arithmetic, cross-join baseline
Q9	Does book size hurt retention?	Conditional aggregation with confound controls
Q10	Campaign funnel by objective	Nested aggregate windows, funnel rates
Q11	Growth opportunity matrix	Window benchmarks, CASE-based classification
Q12	Churn-risk watchlist	Period comparison, materiality thresholds

Two things the queries do deliberately

- **Every denominator is NULLIF-guarded.** A campaign-day with impressions and zero clicks is normal; an unguarded CTR divides by zero somewhere in 195,000 rows.
- **Confounds are handled, not ignored.** Q9's book-size finding would be meaningless without controlling for tier mix, and Q4's cohort denominator is taken from the advertiser dimension rather than from month-0 spenders — the shortcut version reports retention above 100%.

Engine

DuckDB — zero-config, reads CSV natively, and its SQL dialect is close enough to Postgres / Snowflake / BigQuery to port with minor changes. DuckDB-specific syntax (`QUALIFY`, `read_csv_auto`) is flagged in comments wherever it appears.

Appendix — on the data, honestly

The dataset is synthetic, and deliberately so: advertiser-level revenue is commercially confidential and never public. The generator builds in six business patterns for the analysis to find — Q4 seasonality, a segment-level demand shock, tier-differentiated churn, a sales-rep ramp curve, a book-size retention penalty, and an under-invested high-return vertical — rather than producing random noise.

The part worth reading

Getting those patterns to survive into the data took more work than generating it did. **Two rounds of the generator produced statistically dead findings** — a segment too thin to carry a trend, and a book-size effect resting on a two-rep comparison. Both were diagnosed and fixed rather than written up as results. The reasoning is documented in the generator's comments, because it is the part of the process that matters most: **an analysis is only as trustworthy as the check that its finding is real.**

What this analysis cannot tell you

- **The data is synthetic.** It demonstrates method rather than describing a real market. No conclusion here is a claim about any actual company.
- **ROAS is attributed, not incremental.** Every return figure inherits whatever attribution model produced it. Incrementality requires a holdout test, not a query.
- **The declines are described, not explained.** The analysis establishes *that* SEA Gaming contracted and *how much* it cost. Whether the cause is competitive, regulatory or seasonal is not answerable from spend data alone — which is why recommendation 1 is a review, not a fix.
- **Churn is defined as an account closing.** Advertisers who cut spend by 90% but stay open are counted as retained. Q12 partly covers that gap; the tier churn rates in §2 read the strict definition.

Reproducing it

Step	Command
1. Install (Python 3.11+)	<code>pip install -r requirements.txt</code>
2. Regenerate the dataset (optional)	<code>python data/generate_data.py</code>
3. Build DuckDB, run 12 queries, export extracts	<code>python analysis/run_analysis.py</code>
4. Render the dashboard	<code>python analysis/build_dashboard.py</code>
5. Rebuild this PDF	<code>python portfolio/build_ads_report.py</code>

All randomness is seeded (SEED = 20260218), so every number in this document reproduces exactly.

Yizhou Liu · github.com/liuyizhou0402/data-analysis · Full source, SQL, data generator and Tableau build guide in the repository.